

Filter Wizard

Recuperatorio parcial

Created on 06/23/2025

**ANALOG
DEVICES**

Created on 06/23/2025



Filter Wizard Design Report

Filter Requirements for Low-Pass, 4th order Chebyshev

Specifications: Optimize: Power; +Vs: 5; -Vs: -5

Gain: 0 dB

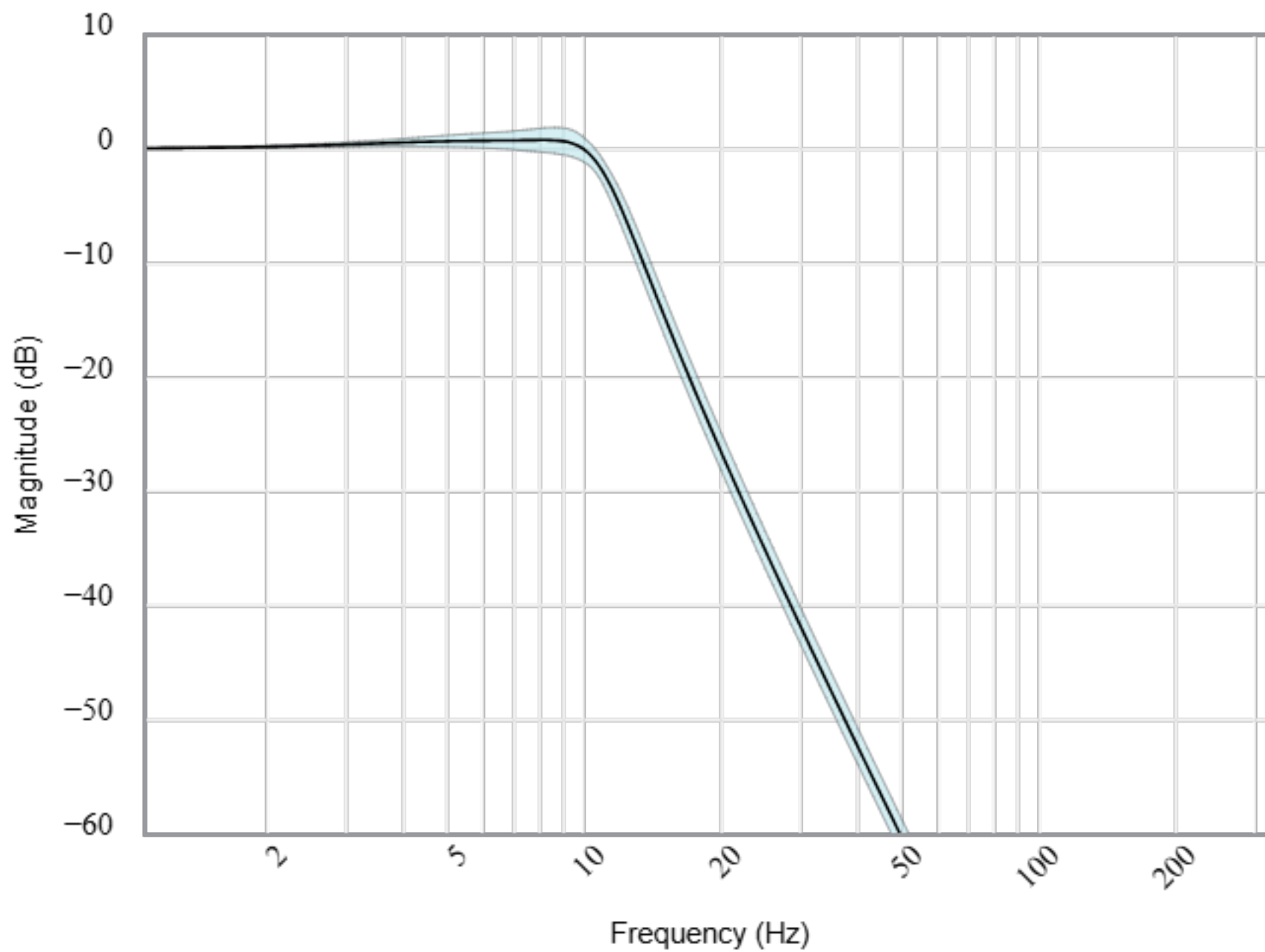
Passband: -3dB at 11Hz

Stopband: -40dB at 33Hz

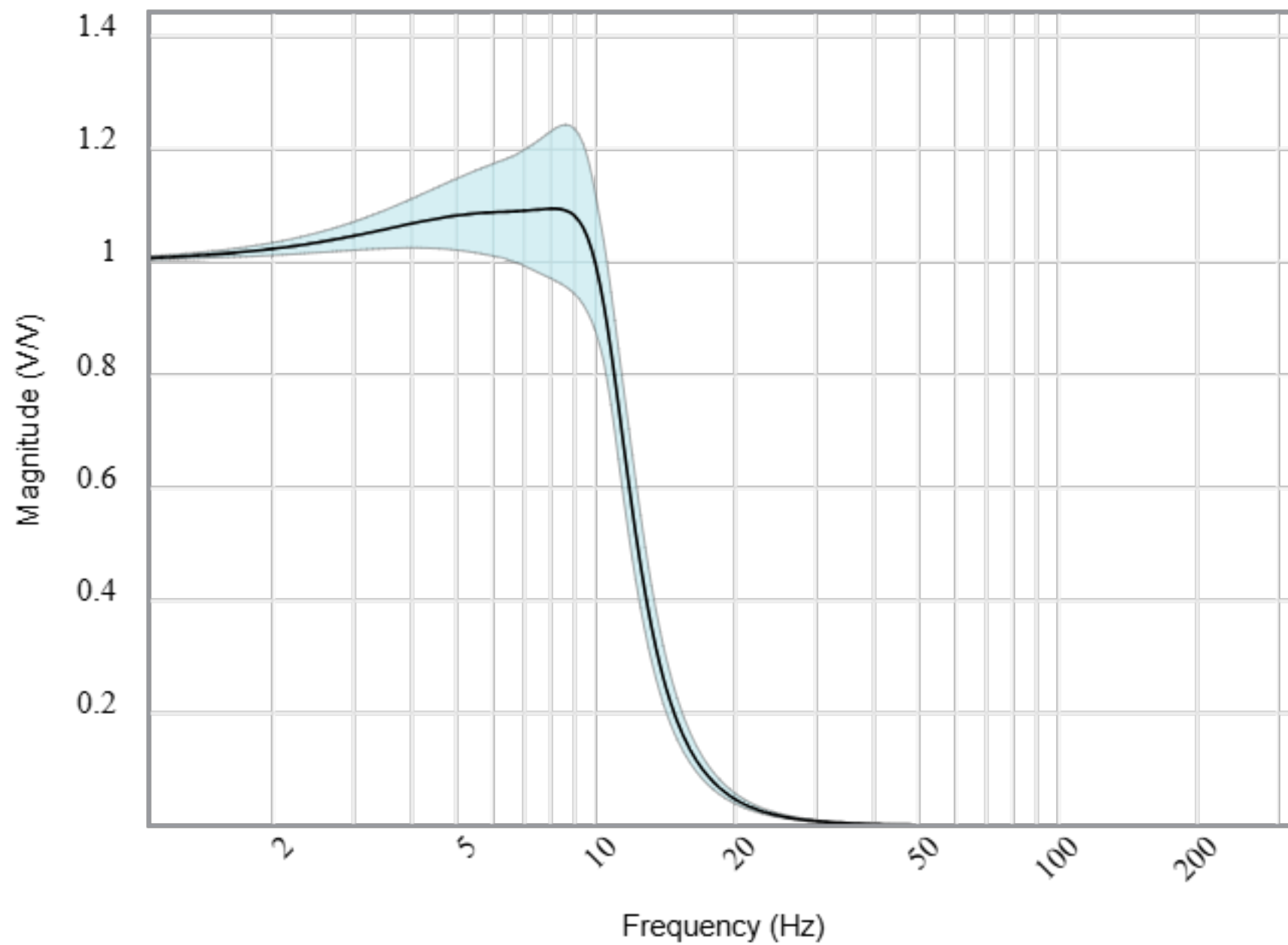
Component Tolerances: Capacitor = 1%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

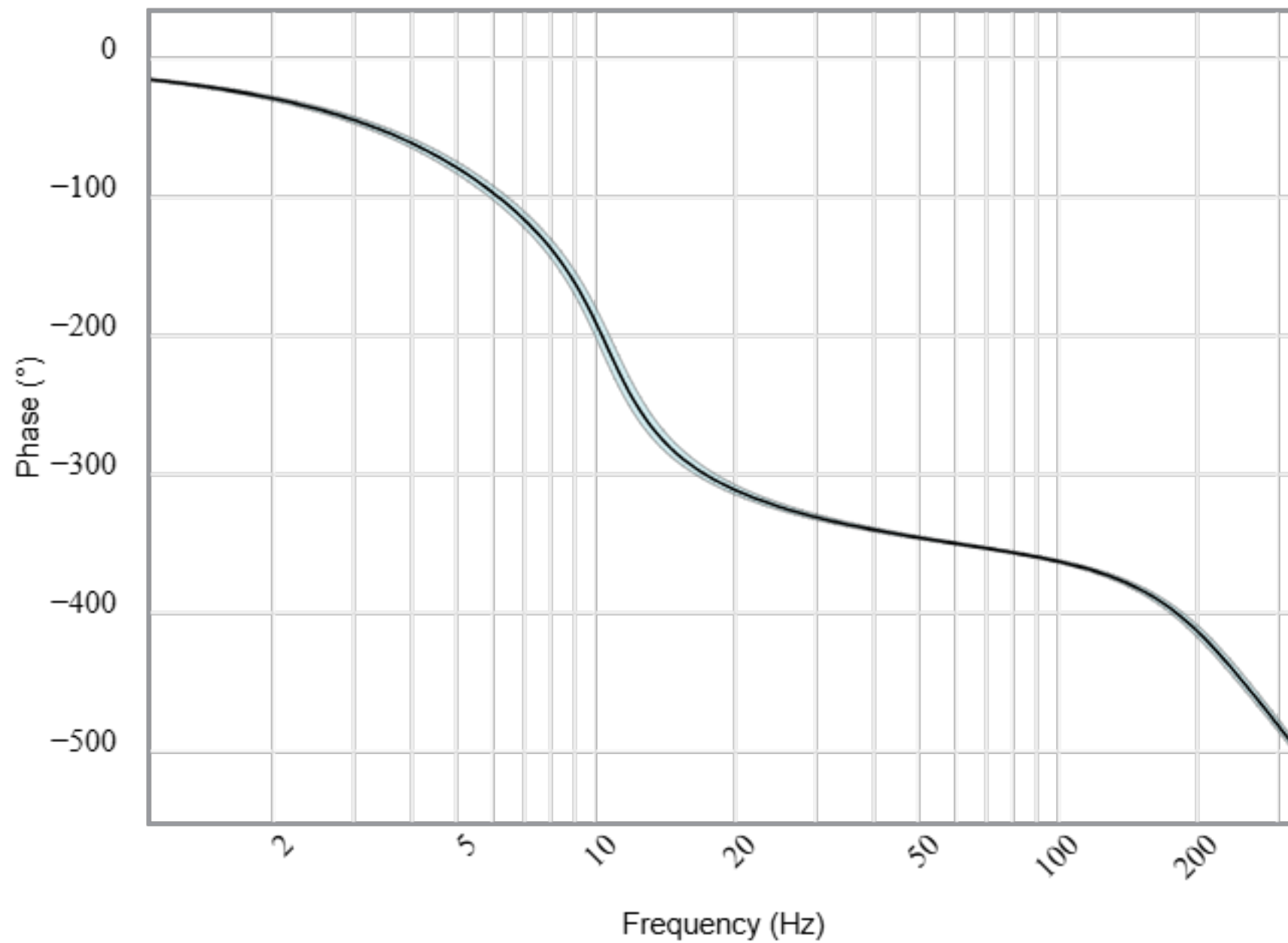
Magnitude(dB)



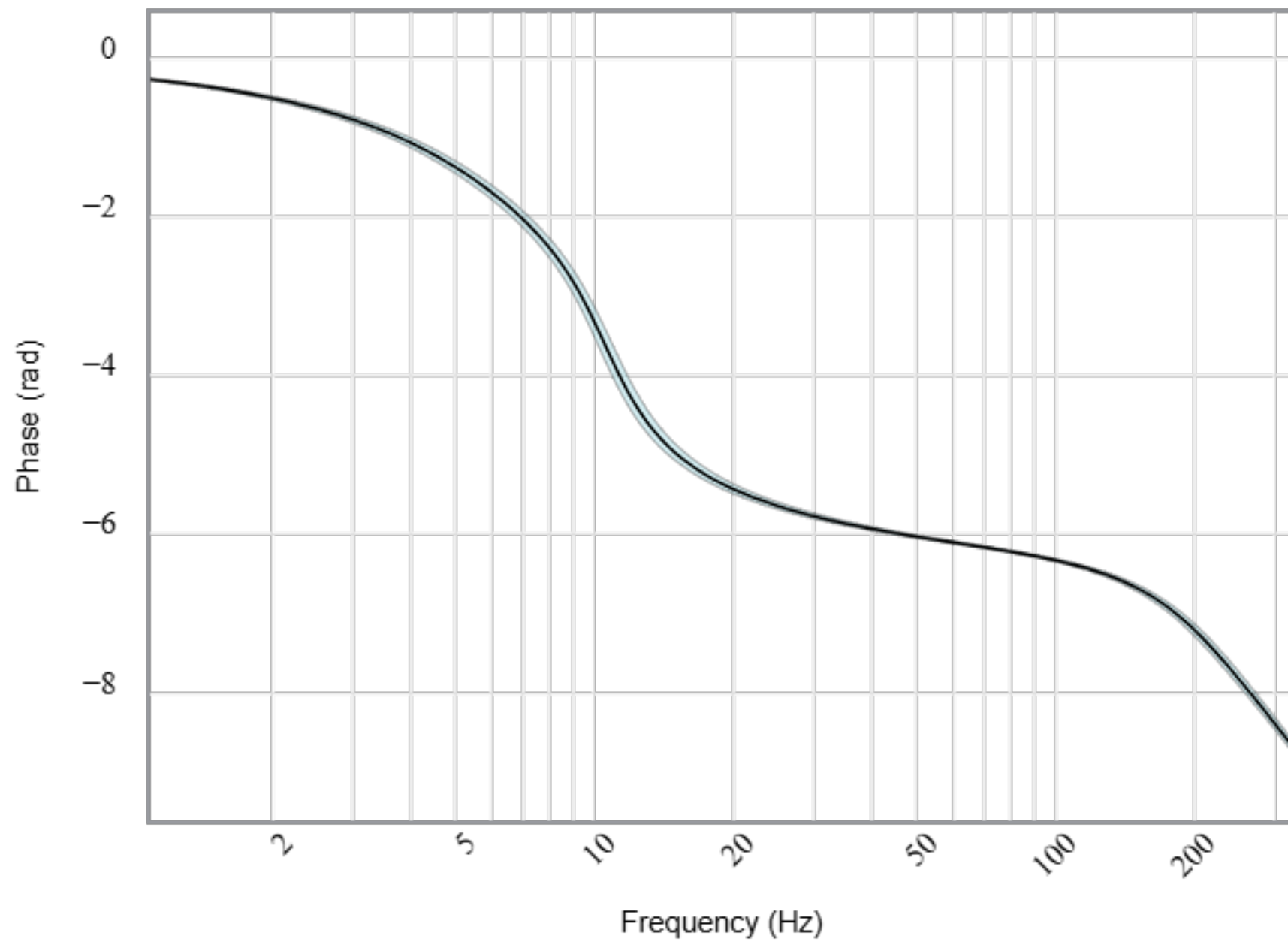
Magnitude(Volts per Volt)



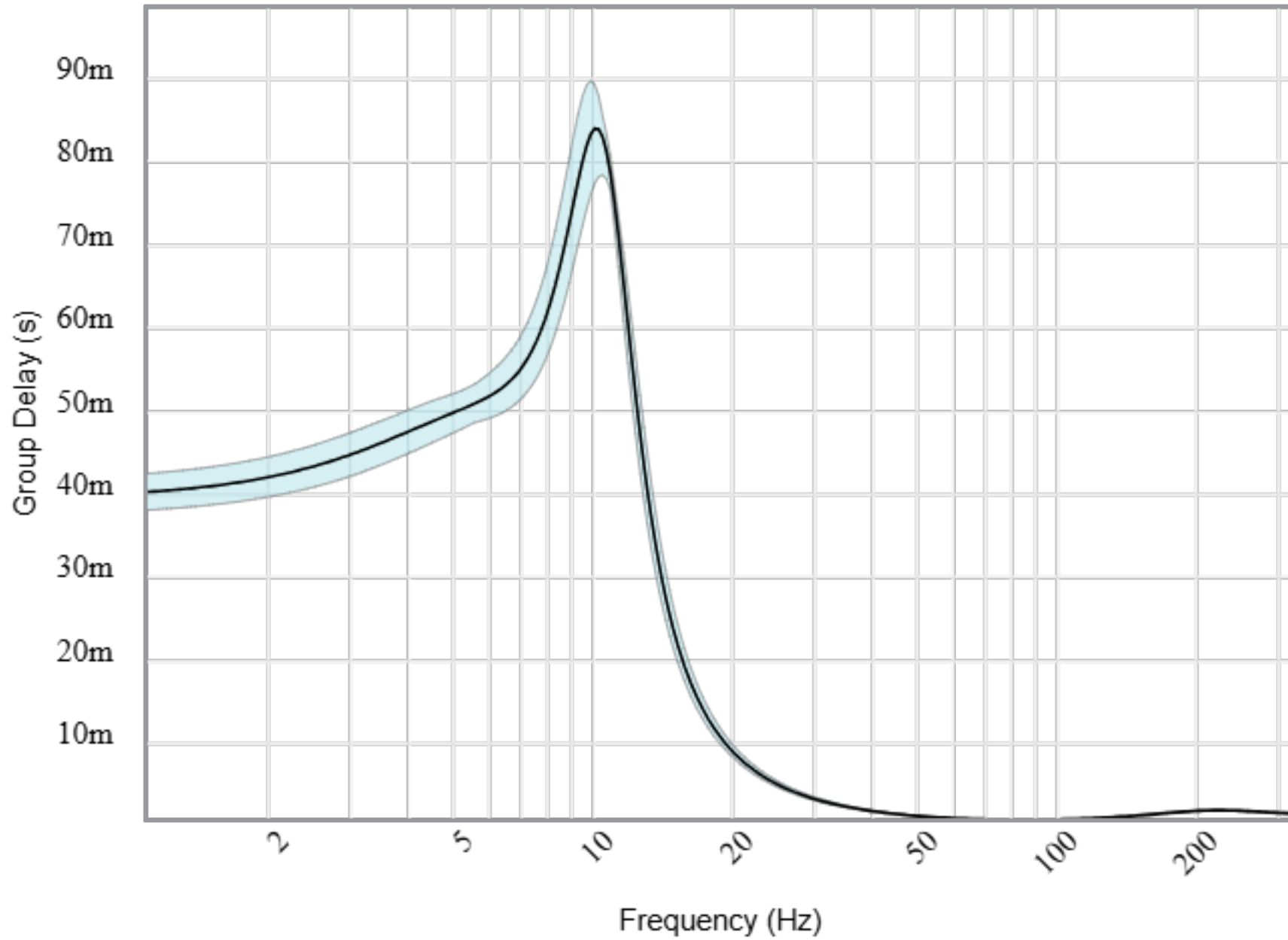
Phase(degrees)



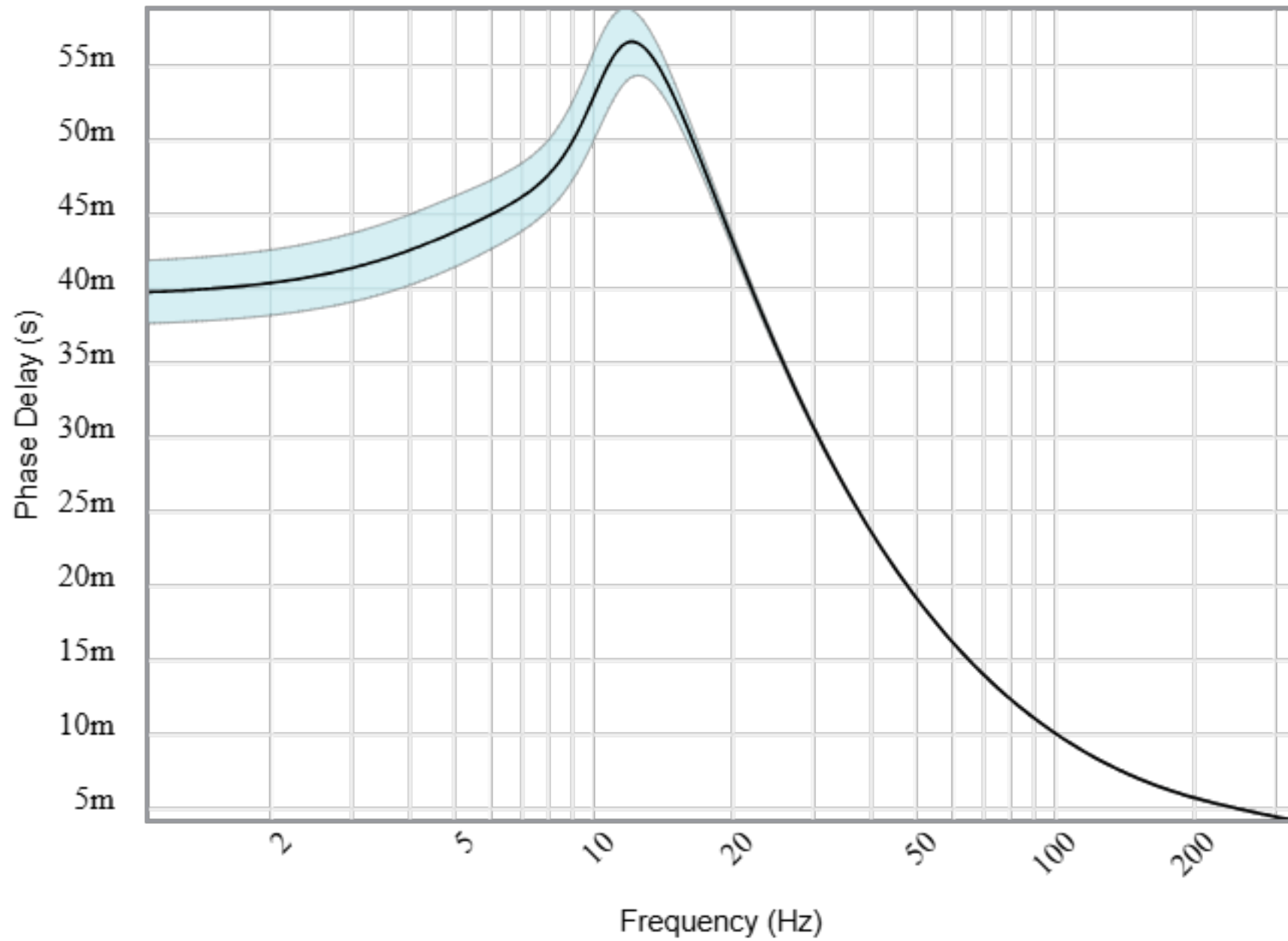
Phase(radians)



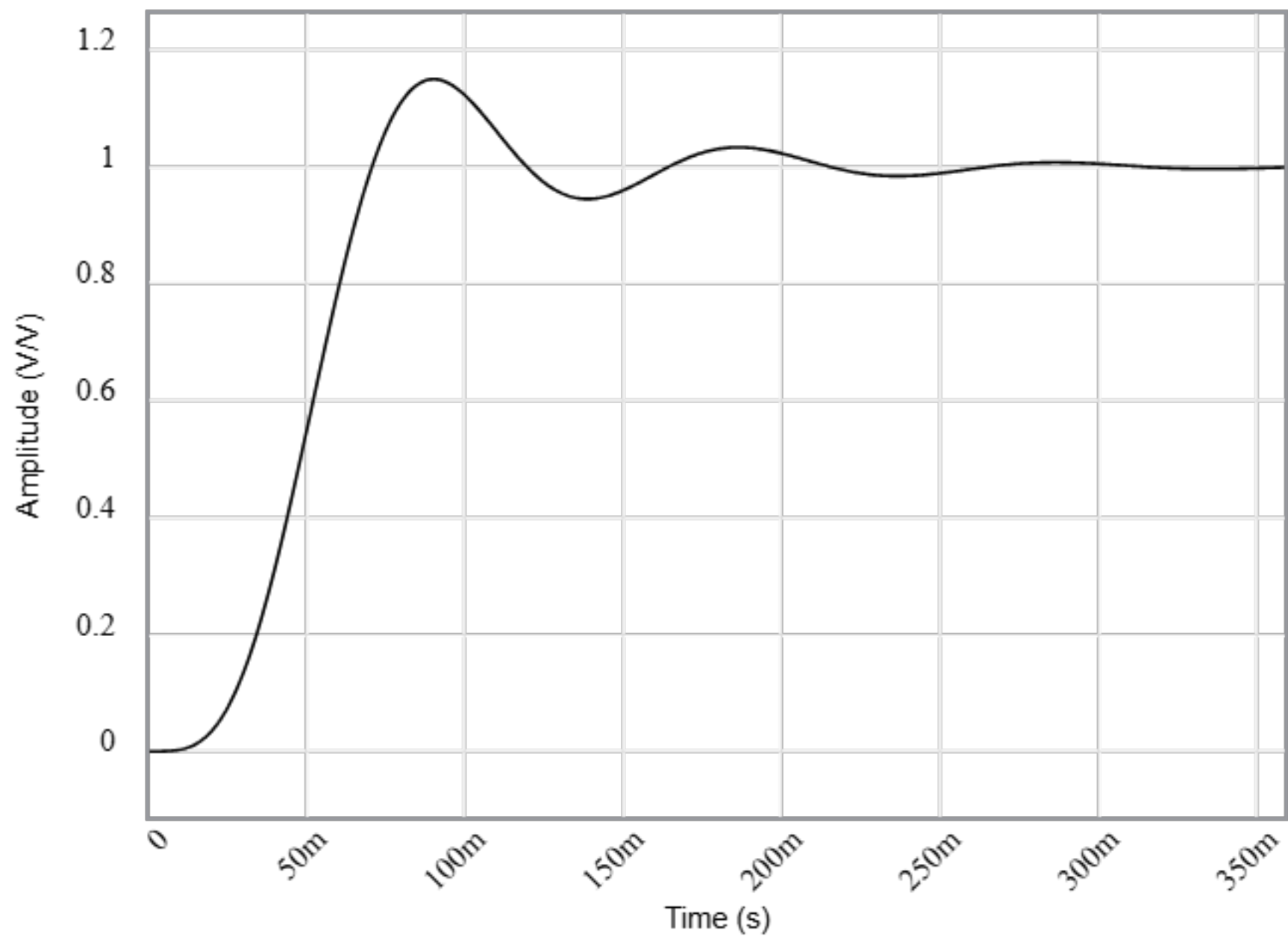
Group Delay



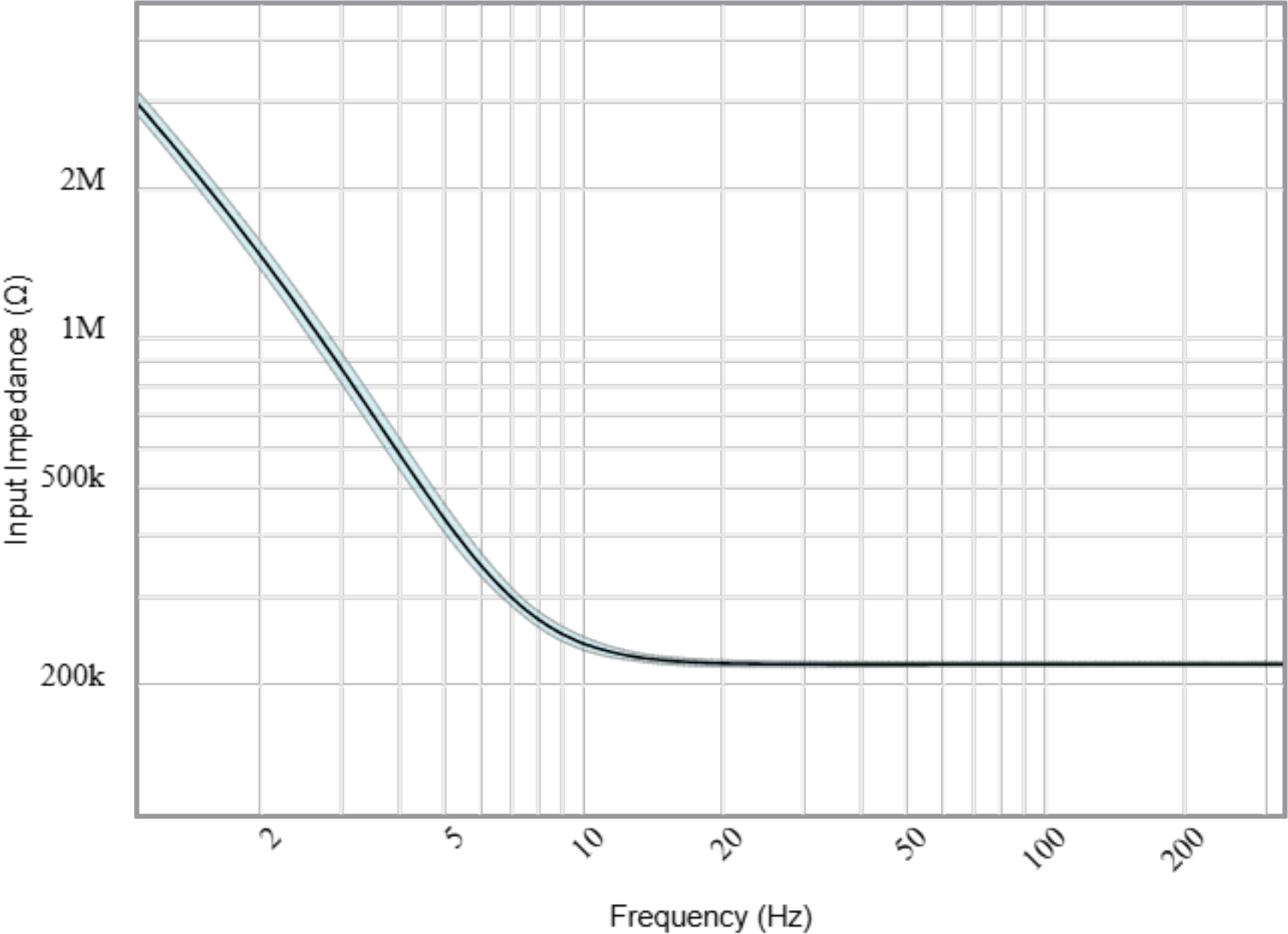
Phase Delay



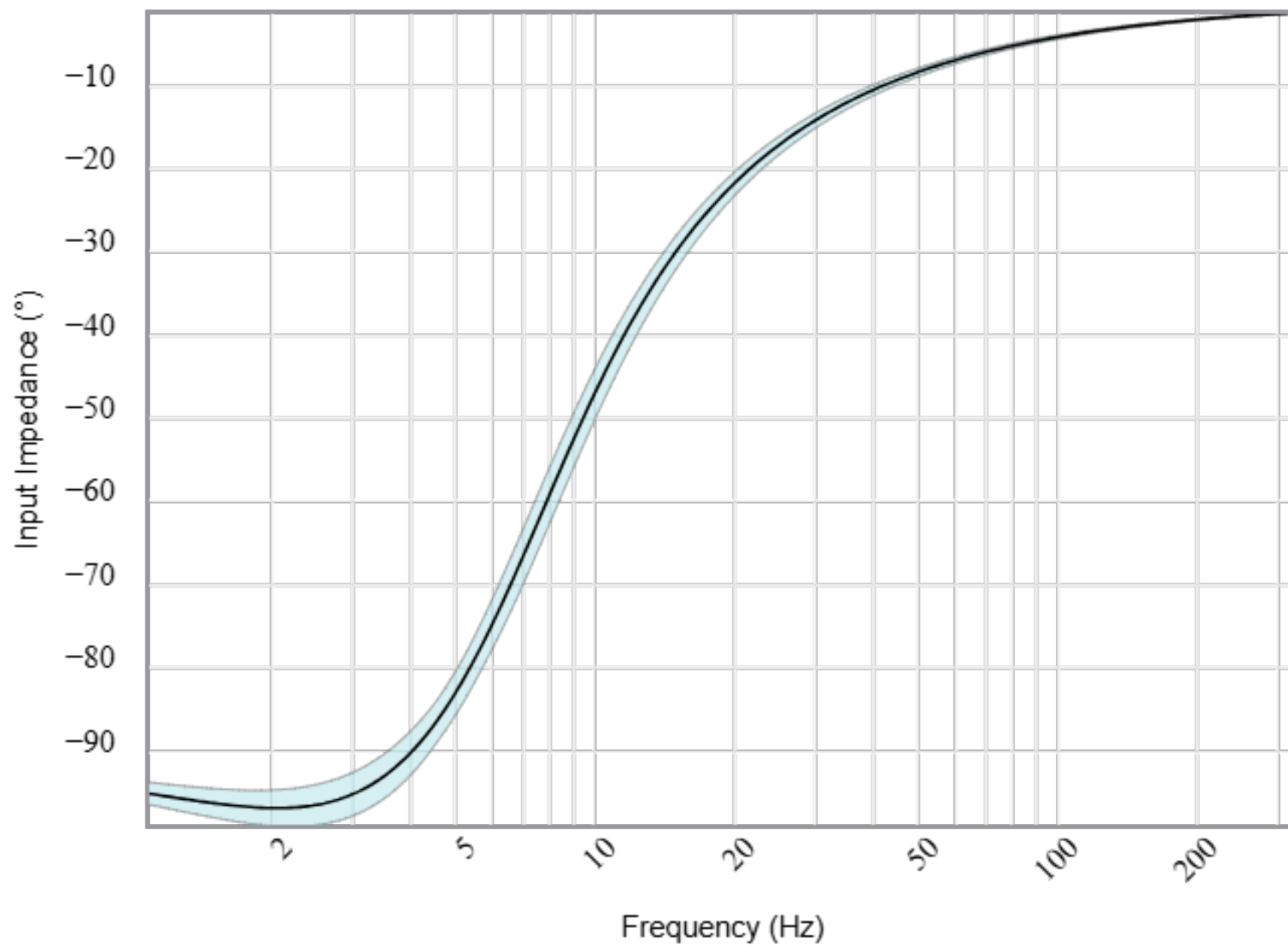
Step Response



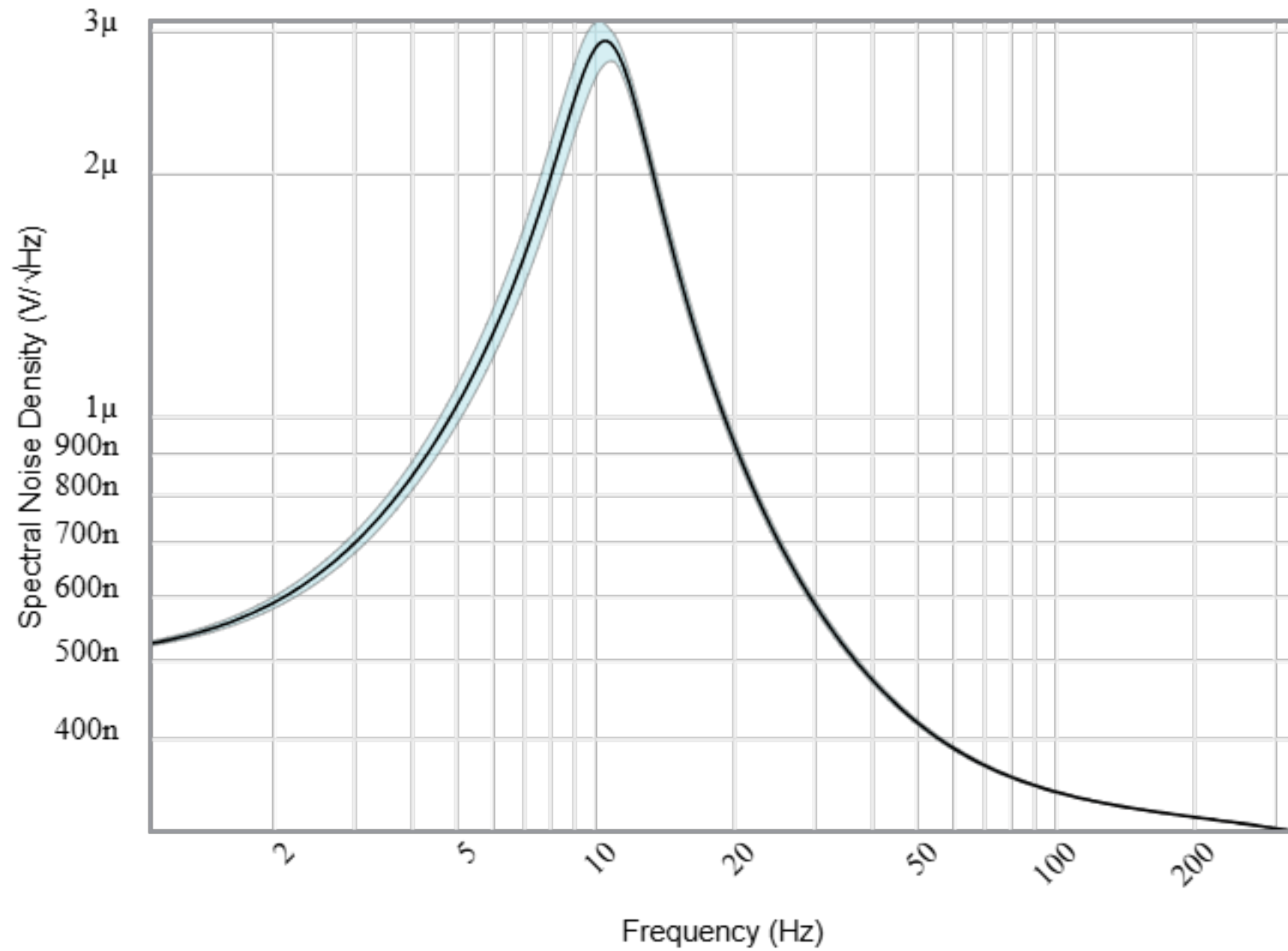
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 2 op amp stage(s) with the following characteristics



2nd order
Low-Pass
Sallen Key

Target

Simulated

Gain (V/V):

1

1 to 1

f_p (Hz):

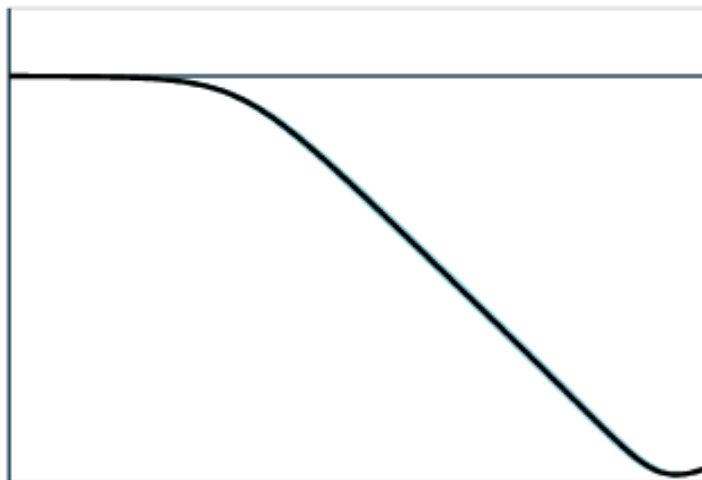
7.14

6.79 to 7.66

Q:

619m

620m to 719m



2nd order
Low-Pass
Sallen Key

Target

Simulated

1

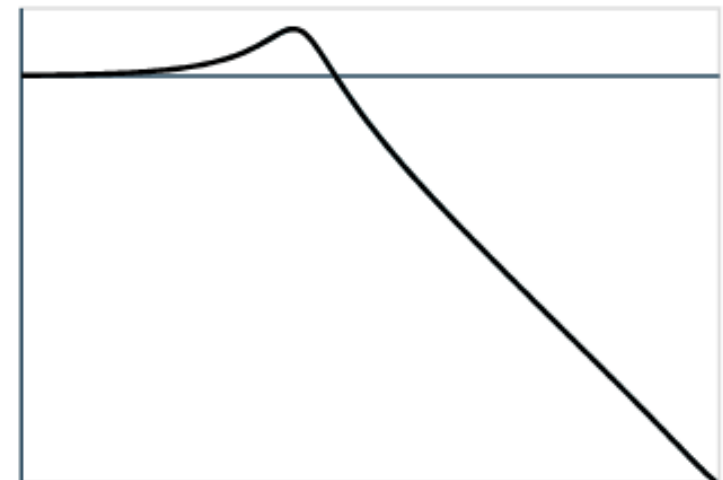
1 to 1

10.4

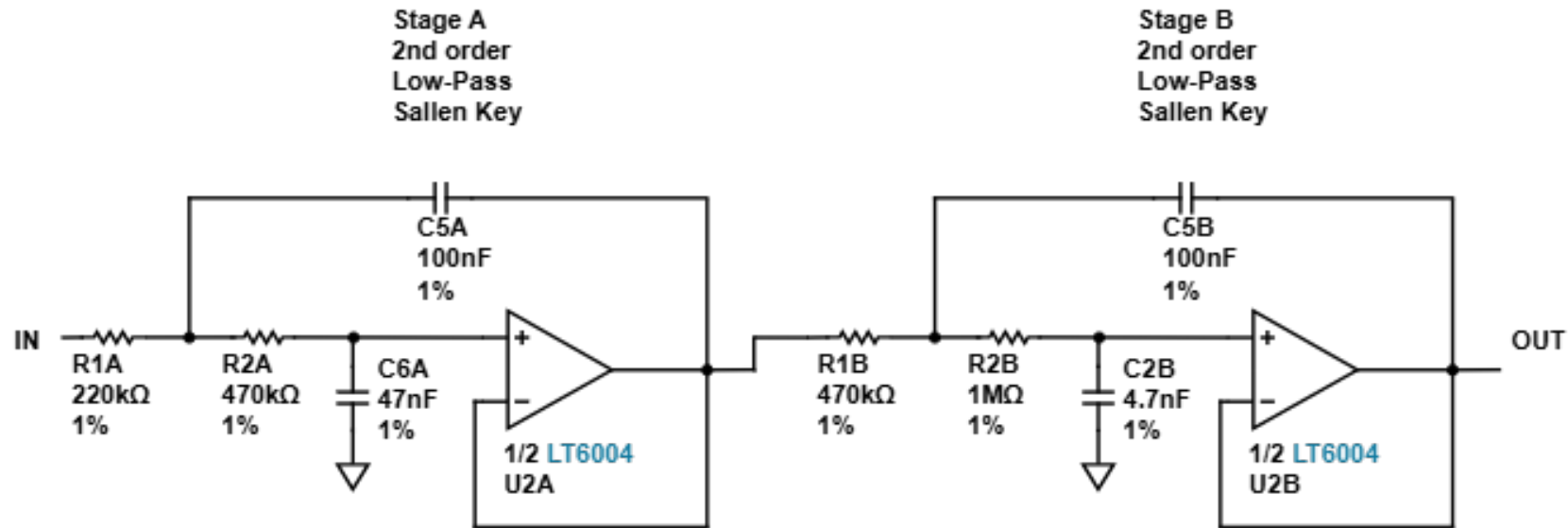
10.3 to 10.9

2.18

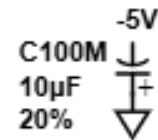
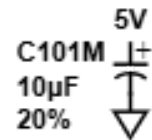
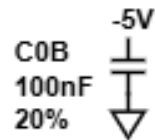
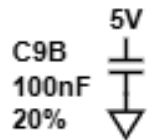
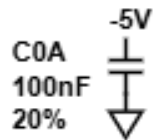
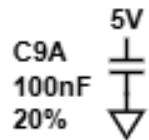
2.1 to 2.23



Circuit



BYPASS CAPACITORS



SPARES Why The Spares?

